2K20/EE/231

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V SEMESTER

B.Tech

END TERM EXAMINATION ***Nov/Dec-2022***

HU-307 BASIC ECONOMETRICS

Time: 03:00 Hours *Max. Marks: 50*

Note : All questions carry equal marks.
Assume suitable missing data, if any.

1. What do you mean by Econometrics? Explain its methodology. Also discuss its significance for engineering students. [CO-1][10] 4.3

2. You have been asked to find out whether effect of COVID-19 on engineers of different branches are significantly different from each other. Engineering branches have been grouped as Traditional, Modern and Digital. What size of sample and which econometrics method you will use to study? Discuss in detail with taking a hypothetical case. [CO3][10]

3. a) Elucidate Ordinary Least square (OLS) method. How are OLS estimates different from the estimates of Method of Maximum Likelihood (ML)? [CO-2][5] 5

b) Explain: i) Gauss Markov Theorem ii) Monte Carlo Experiments [CO-2][5]

4. a) Your data set has Heteroscedasticity. What does it mean and how you will correct it before analysis? 6

[CO-2] [5]

b) What is the significance of Coefficient of Determination and how is it different from Coefficient of Correlation? (CO-1) [5]

5. a). Why does the Normal Distribution hold the most honourable position in Probability theory? When does Binomial or Poison

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distribution tend to a Normal distribution? [CO-3] [5]

b) Consider the following distribution; [CO-1] [2.5x2=5]

	Distribution A	Distribution B
Mean	100	90
Median	90	80
Standard Deviation	10	10

Find out whether following are true or False. Give reasons:

i. Distribution A has same degree of variation as Distribution B
ii. Both distributions have the same degree of skewness.

6 a) What is F-test and when is it used? [CO-3] [5]

b) A manufacturer of a certain make of bulbs claims that their bulbs have a mean value of 25 months with a standard deviation of 5 months. A random sample of 5 such bulbs have the following values: [CO1] [5]

Life in Months: 24, 26, 30, 20, 18.

Can you regard the manufacturers claim to be valid at 1% level of significance? The table values of appropriate statistics at the said level are 4.032, 3.707 and 3.499 at 5,6, and 7 degrees of freedom.

7. Differentiate between any two of the following and discuss their significance in taking decision. [CO-1] [5x2=10]

a) One-tailed test vs. Two tailed test

b) Qualitative data vs Quantitative data.

c). Type 1 error vs. Type II error

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